

Resilient Constrained Consensus and Distributed Optimization: An Integrated Exposition and Open-Source Implementation

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Abstract—We present an integrated exposition of two closely-related Byzantine-resilient algorithms by Zhu, Lin, Velasquez, and Liu. *Part I* treats the resilient *constrained consensus* problem on complete graphs: n agents, each holding a state confined to a private convex set $\mathcal{X}_i$, must reach consensus at a common point in $\mathcal{X} = \bigcap_{i \in \mathcal{H}} \mathcal{X}_i$ despite up to f Byzantine agents that may broadcast arbitrary, possibly *recipient-dependent*, falsified values. The protocol is a single-equation update with closed-form analytic contraction rate $\rho = 1 - (\mu^2 k - 4f - 2f\mu^2 + \mu^2)\alpha + 4|\mathcal{H}|^3 \alpha^2 \in (0, 1)$ for honest-agent Lyapunov $V(t) = \sum_{i \in \mathcal{H}} \|x_i(t) - x^*\|^2$. *Part II* treats the resilient *distributed optimization* problem on general directed graphs: n agents must minimize $f(x) = \sum_{i \in \mathcal{V}} f_i(x)$ despite up to β Byzantine in-neighbors per agent. The protocol is a Tverberg-style subgradient method that aggregates neighbors via a point in the intersection of convex hulls of objective subsets. We provide closed-form constructions for $d = 1$ (any β) – in which case the construction reduces to a trimmed mean – and for $d = 2, \beta = 1$ (Tverberg point of four points via crossing segments). Both algorithms are accompanied by paper-faithful open-source implementations.

Index Terms—Resilient consensus, constrained consensus, distributed optimization, Byzantine adversaries, feasibility redundancy, Tverberg theorem, subgradient method, exponential consensus.

I. INTRODUCTION

BYZANTINE-resilient algorithms are a foundational primitive in modern multi-agent systems, with applications spanning networked control, distributed sensor fusion, decentralized optimization, and federated machine learning [4], [5], [13]. This exposition examines two complementary problems in this space, both introduced by Zhu, Lin, Velasquez, and Liu in successive years of the American Control Conference, hereafter ACC22 [1] and ACC23 [2].

Part I – Constrained consensus (ACC22): Constrained consensus asks n agents to reach a common state while honoring private convex constraints [6]–[8]. In a multi-robot

motion-planning scenario, each agent’s $\mathcal{X}_i$ encodes its local safety envelope, and consensus must lie in the intersection $\mathcal{X} = \bigcap_i \mathcal{X}_i$. When some agents are Byzantine and may transmit arbitrary – and possibly *recipient-dependent* – falsified values, the standard projected-consensus dynamics fail. The ACC22 algorithm is the first study of resilient discrete-time constrained consensus, focusing on the analytically tractable setting of complete communication graphs.

Part II – Distributed optimization (ACC23): Distributed optimization asks n agents to cooperatively minimize $f(x) = \sum_{i \in \mathcal{V}} f_i(x)$, where each f_i is private to agent i . Mean-subsequence-reduced (MSR) algorithms [14], [15] guarantee *approximate* resilience – iterates end up in a bounded neighborhood of the optimum. Detection-based algorithms achieve full resilience on complete graphs [16]. The ACC23 algorithm closes the gap for general directed graphs by combining *objective redundancy* [17] with a *Tverberg-style* aggregator [10], [18] that excludes adversarial inputs by Tverberg’s theorem.

Why combine the two: The two papers occupy complementary niches: ACC22 addresses an inherently geometric question (intersection of convex sets) on a topologically simple network, while ACC23 addresses an inherently optimization question (joint minimum of convex functions) on a topologically rich network. The two share notation conventions, the Byzantine threat model, and a common author group; they reference each other’s machinery implicitly (e.g., the running-sum primitive used in distributed averaging). Reading them together gives the reader a coherent map of the resilient-consensus landscape circa 2022–2023.

A. About this exposition

The algorithms, theorems, and proofs presented here are due to the source papers. This document contributes: (i) a *unified* treatment with shared preliminaries and consistent notation across the two parts; (ii) explicit pseudocode for every algorithm and parameter checker; (iii) closed-form derivations of the two-Tverberg construction in $\mathbb{R}^2$; (iv) paired open-source Python and MATLAB implementations (ACC22.py, ACC22.m, ACC23.py, ACC23.m) that reproduce the numerical experiments of both papers and support their worst-case Byzantine attack classes.

This document is an integrated exposition adapted from two related publications by J. Zhu, Y. Lin, A. Velasquez, and J. Liu: “Resilient Constrained Consensus over Complete Graphs via Feasibility Redundancy” (*Proc. ACC 2022*, pp. 3418–3422), and “Resilient Distributed Optimization” (*Proc. ACC 2023*, pp. 1307–1312). The contributions of this exposition are (i) a unified treatment of the two papers, with a shared preliminaries section and consistent notation; (ii) self-contained re-derivations with full pseudocode and explicit parameter checkers; and (iii) paired open-source implementations in Python (ACC22.py, ACC23.py) and MATLAB (ACC22.m, ACC23.m) that reproduce the numerical examples of both papers. The original algorithms and theorems are due to Zhu, Lin, Velasquez, and Liu.

B. Outline

Section II presents shared preliminaries: graph notation, projection onto convex sets, the subgradient method, and the Byzantine threat model. Section III (Part I) presents the resilient constrained consensus algorithm of ACC22, the necessary $2f$ -redundancy (Theorem 1), and the sufficient k -redundancy with analytic contraction rate (Theorem 2). Section IV (Part II) presents the resilient distributed optimization algorithm of ACC23, the Tverberg-style aggregator with closed forms, and the asymptotic and partial-rate convergence theorems. Section V reports numerical experiments produced by the open-source implementations. Section VI concludes.

II. SHARED PRELIMINARIES

A. Graph notation

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed graph on the vertex set $\mathcal{V} = \{1, \dots, n\}$ with edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$. We adopt the convention that $(j, i) \in \mathcal{E}$ means j is an in-neighbor of i , written $j \in \mathcal{N}_i^-$. The set of out-neighbors is $\mathcal{N}_i^+ = \{j : (i, j) \in \mathcal{E}\}$ and we write $d_i^- = |\mathcal{N}_i^-|$, $d_i^+ = |\mathcal{N}_i^+|$. An undirected edge is a pair of opposite-direction directed edges. A directed graph is *strongly connected* if every vertex is reachable from every other, and an undirected graph is k' -connected if no removal of $k' - 1$ vertices disconnects it. The complete graph on n vertices is denoted $\mathcal{K}_n$.

B. Convex projection

For any closed convex $\subseteq \mathbb{R}^m$ and $x \in \mathbb{R}^m$, the Euclidean projection $P[x] = \arg \min_{y \in \mathcal{C}} \|x - y\|$ exists and is unique; we denote $(x, \cdot) = \|x - P[x]\|$. Closed-form projections are available for several special cases used throughout this exposition:

- **Box** = $[\ell, u]$: $P[x] = \text{clip}(x, \ell, u)$.
- **Ball** = $\{y : \|y - c\| \leq r\}$: radial scaling toward c .
- **Halfspace** = $\{y : a^\top y \leq b\}$: a single affine correction $P[x] = x - \frac{(a^\top x - b)_+}{\|a\|^2} a$.

For arbitrary polyhedra = $\{y : Ay \leq b\}$ a quadratic program is required.

C. Subgradient method

For a convex $h : \mathbb{R}^d \rightarrow \mathbb{R}$, a vector $g \in \mathbb{R}^d$ is a *subgradient* at x if $h(y) \geq h(x) + g^\top(y - x)$ for all y . When h is differentiable, $g = \nabla h(x)$. We use $\|\cdot\|$ for the Euclidean 2-norm.

D. Threat model

Partition the n agents into honest $\mathcal{H}$ and Byzantine $\tilde{\mathcal{A}} = \mathcal{V} \setminus \mathcal{H}$. The number of Byzantine agents is bounded either *globally* (f -total: $|\tilde{\mathcal{A}}| \leq f$) or *locally* (β -local: $|\mathcal{N}_i^- \cap \tilde{\mathcal{A}}| \leq \beta$ for every honest i). Part I (ACC22) uses the f -total model, while Part II (ACC23) uses the β -local model.

Definition 1 (Byzantine model, ACC22 footnote 1). A Byzantine agent $j \in \tilde{\mathcal{A}}$ may transmit arbitrary values to other agents and – in the strongest form of the model – may transmit different values $x_{ji}(t) \neq x_{jk}(t)$ to different recipients $i \neq k$ at the same time t .

Algorithm 1 Resilient Constrained Consensus (ACC22 Algorithm 1)

Require: Honest agents share knowledge of $\mathcal{G} = \mathcal{K}_n$, f , α ; each i has private $\mathcal{X}_i$ and initial $x_i(0)$.

- 1: **for** $t = 0, 1, 2, \dots$ **do**
- 2: Each agent i broadcasts $x_i(t)$ (Byzantine: per-recipient).
- 3: **for** each honest $i \in \mathcal{H}$ **do**
- 4: Compute distances $d_{ji}(t) = \|x_{ji}(t) - x_i(t)\|$ for $j \in \mathcal{N}_i^- \setminus \{i\}$.
- 5: Sort, discard the f farthest, retain set $\mathcal{M}_i(t)$.
- 6: Set $x_i(t+1) = P_{\mathcal{X}_i}[x_i(t) + \alpha \sum_{j \in \mathcal{M}_i(t)} (x_{ji}(t) - x_i(t))]$.
- 7: **end for**
- 8: **end for**

The per-recipient dimension distinguishes Byzantine from the broadcast “malicious” model used in some of the literature, in which each adversary must send the same value to all recipients. Our implementations realize the strongest Byzantine model through an (n, n, d) message buffer and a `two_faced` attack class that supplies independently-perturbed extremal values to each recipient.

III. PART I: RESILIENT CONSTRAINED CONSENSUS ON COMPLETE GRAPHS

A. Problem statement

Let $\mathcal{G} = \mathcal{K}_n$ and let each agent $i \in \mathcal{V}$ hold state $x_i(t) \in \mathbb{R}^m$ with private closed convex feasibility set $\mathcal{X}_i$. Assume $\mathcal{X} = \bigcap_{i \in \mathcal{H}} \mathcal{X}_i$ is non-empty; for the convergence results we further require $\mathcal{X} = \{x^*\}$ to be a singleton. The goal is for every honest agent to converge to the common point x^* :

$$\lim_{t \rightarrow \infty} x_i(t) = x^* \quad \forall i \in \mathcal{H}. \quad (1)$$

B. The algorithm

The protocol is a single-equation iterative scheme (ACC22 Eq. (1)):

$$x_i(t+1) = P_{\mathcal{X}_i} \left[x_i(t) + \alpha \sum_{j \in \mathcal{M}_i(t)} (x_{ji}(t) - x_i(t)) \right], \quad i \in \mathcal{H}, \quad (2)$$

where $\alpha > 0$ is a step size and $\mathcal{M}_i(t)$ is the *retain set* obtained by computing distances $d_{ji}(t) = \|x_{ji}(t) - x_i(t)\|$ for every in-neighbor j , sorting, discarding the f neighbors with the largest distance, and retaining the surviving set of $n - 1 - f$ agents.

The update (2) is synchronous: every honest agent reads from a common snapshot of the network state. Our implementation enforces this by copying the state vector at the start of every round and writing back atomically.

C. Necessary redundancy

Definition 2 (k -redundancy of feasibility, ACC22 Def. 1). The honest set $\mathcal{H}$ is k -redundant if for every subset $\mathcal{S} \subseteq \mathcal{H}$ with $|\mathcal{S}| \geq |\mathcal{H}| - k$, $\bigcap_{i \in \mathcal{S}} \mathcal{X}_i = \bigcap_{i \in \mathcal{H}} \mathcal{X}_i$.

Theorem 1 (Necessary condition, ACC22 Thm. 1). *If there are f Byzantine agents, all honest agents can converge to a common point in $\mathcal{X}$ via Algorithm 1 only if $\mathcal{H}$ is $2f$ -redundant.*

Sketch: By contrapositive: if $\mathcal{H}$ is not $2f$ -redundant, there exists $\hat{\mathcal{A}} \subset \mathcal{H}$ with $|\hat{\mathcal{A}}| \geq |\mathcal{H}| - 2f$ and $\bigcap_{i \in \hat{\mathcal{A}}} \mathcal{X}_i \supsetneq \mathcal{X}$, so we can pick $x \in \bigcap_{i \in \hat{\mathcal{A}}} \mathcal{X}_i \setminus \mathcal{X}$. Initialize $x_i(0) = x$ for $i \in \hat{\mathcal{A}}$ and let the f Byzantines broadcast x to every honest agent. Induction on t shows $x_i(t) = x$ for all $i \in \hat{\mathcal{A}}$, $t \geq 0$, contradicting convergence into $\mathcal{X}$ for any honest agent in $\mathcal{H} \setminus \hat{\mathcal{A}}$. ■

D. Sufficient conditions and analytic contraction rate

The sufficient conditions add a regularity constant on the convex sets:

Definition 3 (Set regularity, ACC22 Eq. (2)). *There exists $\mu \in (0, 1]$ such that for all $x \in \mathbb{R}^m$ and every $\mathcal{S} \subseteq \mathcal{H}$ with $|\mathcal{S}| \geq n - k$,*

$$\max_{i \in \mathcal{S}} \mu(x, \mathcal{X}_i) \geq \mu \left(x, \bigcap_{i \in \mathcal{S}} \mathcal{X}_i \right). \quad (3)$$

For polyhedral sets $\mathcal{X}_i = \{x : a_i^\top x \leq b_i\}$, Hoffman's lemma [12] guarantees a constant μ exists.

Theorem 2 (Exponential consensus, ACC22 Thm. 2). *Suppose $\mathcal{X} = \{x^*\}$ is a singleton and $\mathcal{H}$ is k -redundant with regularity constant μ . If*

$$k > \frac{4f}{\mu^2} + 2f - 1, \quad (4)$$

and the step size α satisfies

$$\alpha < \frac{\mu^2 k - 4f - 2f\mu^2 + \mu^2}{4|\mathcal{H}|^3}, \quad (5)$$

then there exists $\rho \in (0, 1)$ for which the honest-agent Lyapunov $V(t) := \sum_{i \in \mathcal{H}} \|x_i(t) - x^\|^2$ satisfies $V(t) \leq \rho^t V(0)$ for all $t \geq 0$, with explicit*

$$\rho = 1 - (\mu^2 k - 4f - 2f\mu^2 + \mu^2) \alpha + 4|\mathcal{H}|^3 \alpha^2. \quad (6)$$

Sketch: Define $V_i(t) = \|P_{\mathcal{X}_i}[x_i(t)] - x^*\|^2$ and use the non-expansive property of Euclidean projection to bound $V_i(t+1) \leq V_i(t) - 2\alpha\phi_i(t) + \alpha^2\psi_i(t)$, where ϕ_i collects inner products with the residual $\sum_{j \in \mathcal{M}_i(t)} (x_{ji} - x_i)$ and ψ_i is the squared norm of that residual. Two key lemmas (ACC22 Lem. 3, Lem. 4) bound $\sum \phi_i \geq \frac{\mu^2}{2}(k - \frac{4f+2f\mu^2-\mu^2}{2})V(t)$ and $\sum \psi_i \leq 4|\mathcal{H}|^3 V(t)$. Combining yields $V(t+1) \leq \rho V(t)$ with ρ as in (6); the bound (5) ensures $\rho < 1$. ■

Corollary 1 (Polyhedral case, ACC22 Cor. 1). *If $\mathcal{X}_i = \{x : a_i^\top x \leq b_i\}$ for all $i \in \mathcal{H}$ and $\bigcap_{i \in \mathcal{H}} \mathcal{X}_i = \{x^*\}$, then a regularity constant μ exists by Hoffman's lemma, and the conclusion of Theorem 2 holds.*

IV. PART II: RESILIENT DISTRIBUTED OPTIMIZATION ON DIRECTED GRAPHS

A. Problem statement

Each agent $i \in \mathcal{V}$ has a private convex (not necessarily differentiable) objective $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$. The honest agents must cooperatively minimize $f_{\mathcal{H}}(x) = \sum_{i \in \mathcal{H}} f_i(x)$. Throughout, the adversary set $\hat{\mathcal{A}} \subseteq \mathcal{V}$ is β -local: every honest agent has at most β adversarial in-neighbors.

Algorithm 2 Resilient Distributed Subgradient (ACC23 Algorithm)

Require: Each i has $f_i, \mathcal{N}_i^-, x_i[0]$, step size $\alpha[k]$.

```

1: for  $k = 0, 1, 2, \dots$  do
2:   Each agent  $i$  (honest) broadcasts  $x_i[k]$  to  $\mathcal{N}_i^+$ .
3:   for each  $i \in \mathcal{H}$ , each  $A_{ij} \in \mathfrak{A}_i$  do
4:     Choose  $y_{ij}[k] \in \bigcap_{S \in \mathfrak{B}_{ij}} \text{conv}\{x_h[k] : h \in S\}$ .
5:   end for
6:   Form  $v_i[k] = \frac{1}{1+a_i} \left( x_i[k] + \sum_{j=1}^{a_i} y_{ij}[k] \right)$  with  $a_i = |\mathfrak{A}_i|$ .
7:   Subgradient step:  $x_i[k+1] = v_i[k] - \alpha[k] g_i(v_i[k])$ .
8: end for
```

B. Objective and graph redundancy

The honest agents' iterates must converge to an argmin of $f_{\mathcal{H}}$, but the identity of $\mathcal{H}$ is itself unknown. The path is to require that small subsets of objectives have the *same* argmin as the full set.

Definition 4 (k -redundant network, ACC23 Def. 1). *A network is k -redundant if for every two subsets $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathcal{V}$ of size $n-k$,*

$$x \sum_{i \in \mathcal{S}_1} f_i(x) = x \sum_{i \in \mathcal{S}_2} f_i(x). \quad (7)$$

Lemma 1 (ACC23 Lem. 1). *If a network is k -redundant, then for any $\mathcal{S} \subseteq \mathcal{V}$ with $|\mathcal{S}| = n - k$ and any $\mathcal{L} \supseteq \mathcal{S}$, $x \sum_{i \in \mathcal{S}} f_i(x) = x \sum_{i \in \mathcal{L}} f_i(x)$.*

Corollary 2 (ACC23 Cor. 1). *If a network is k -redundant, then for every $\mathcal{S} \subseteq \mathcal{V}$ with $|\mathcal{S}| \geq n - k$, $x \sum_{i \in \mathcal{S}} f_i(x) = \mathcal{X}^*$, the optimum set.*

Definition 5 ((r, s) -reduced graph, ACC23 Def. 2). *An (r, s) -reduced graph of $\mathcal{G}$ is any digraph obtained by: (i) removing r vertices to leave the induced subgraph on $|\mathcal{S}| = n - r$, and (ii) removing arbitrarily up to s incoming edges from each vertex. $\mathcal{G}$ is (r, s) -resilient if every (r, s) -reduced graph contains a root vertex.*

Lemma 2 (Minimum neighborhood size, ACC23 Lem. 2). *If $\mathcal{G}$ is (r, s) -resilient, then every vertex has at least $r + s + 1$ neighbors.*

For an (r, s) -resilient $\mathcal{G}$, $\kappa_{r,s}(\mathcal{G})$ denotes the minimum, over all (r, s) -reduced graphs, of the number of root vertices. The implementation includes a brute-force computation of $\kappa_{r,s}(\mathcal{G})$ for $n \leq 6$ via exhaustive enumeration.

C. The Tverberg-style aggregator

For each honest agent i , let $\mathfrak{A}_i$ denote the collection of all $(d+1)\beta+1$ -subsets of $\mathcal{N}_i^-$. For each $A_{ij} \in \mathfrak{A}_i$, let $\mathfrak{B}_{ij}$ denote the collection of $(d\beta+1)$ -subsets of A_{ij} .

The non-emptiness of $\bigcap_{S \in \mathfrak{B}_{ij}} \text{conv}\{x_h : h \in S\}$ follows from Tverberg's theorem applied to A_{ij} .

D. Closed form for $d = 1$

For $d = 1$ the convex hull of a $(d\beta+1) = (\beta+1)$ -subset of scalars is just the closed interval between the smallest and

largest of the subset. The intersection over all $\binom{2\beta+1}{\beta+1}$ such intervals of any $(2\beta+1)$ -subset A_{ij} is the closed interval between the $(\beta+1)$ -th smallest and the $(\beta+1)$ -th largest value – the *trimmed-mean range*. The midpoint of this range is a canonical choice for y_{ij} . This is precisely the trimmed-mean reduction the source paper notes for the scalar case.

E. Closed form for $d = 2, \beta = 1$

For $d = 2, \beta = 1, |A_{ij}| = 4$ and $\mathfrak{B}_{ij}$ consists of the four 3-subsets. Tverberg’s theorem with $r = 2, d = 2$ states that any $(r-1)(d+1)+1 = 4$ points in $\mathbb{R}^2$ admit a partition into $r = 2$ groups whose convex hulls intersect. In general position, exactly one of the three 2-2 partitions of four points has crossing segments; the intersection of those two segments is the unique *Tverberg point* of the four-point set, and this point lies in every 3-subset’s convex hull. Computing it requires solving a single 2×2 linear system.

F. General case $d \geq 2, \beta \geq 2$

For larger d or β the construction requires intersecting many convex hulls in $\mathbb{R}^d$, which is computationally expensive (the source paper explicitly flags this in its concluding section). Our implementation raises a clean error in this regime and recommends the comparison aggregators below.

G. Comparison aggregators

The implementation also supports four standard robust aggregators that operate on the raw neighbor values: distance trim, coordinate-wise trimmed mean (CWTM) [20], Krum [21], and the geometric median via Weiszfeld’s iteration. These are not the source paper’s algorithm but are useful for empirical comparison.

H. Convergence

Assumption 1 (Standard subgradient assumptions, ACC23 Asm. 1–3). (i) $\mathcal{X}^*$ has nonempty interior; (ii) all subgradients are uniformly bounded; (iii) the step-size sequence $\{\alpha[k]\}$ is positive, non-increasing, with $\sum \alpha[k] = \infty$ and $\sum \alpha[k]^2 < \infty$.

Theorem 3 (Asymptotic resilience, ACC23 Thm. 1). *Under Asm. 1, if $\mathcal{G}$ is $(\beta, d\beta)$ -resilient and the network is $(n - \kappa_{\beta, d\beta}(\mathcal{G}))$ -redundant, then every honest agent’s iterates $x_i[k]$ converge to a common point in $\mathcal{X}^*$.*

Theorem 4 (Partial $\mathcal{O}(1/\sqrt{T})$ rate, ACC23 Thm. 2). *Under Asm. 1 and 2, if $\mathcal{G}$ is $(\beta, d\beta)$ -resilient, the network is $(n - \kappa_{\beta, d\beta}(\mathcal{G}))$ -redundant, and $\alpha[k] = 1/\sqrt{T}$ for $k \in \{0, \dots, T-1\}$, then there exists a subset $\mathcal{S} \subseteq \mathcal{H}$ with $|\mathcal{S}| \geq \kappa_{\beta, d\beta}(\mathcal{G})$, a constant $C \geq 1$, and a sub-sequence $\mathcal{T} \subseteq \{0, \dots, T-1\}$ with $|\mathcal{T}| \geq T/C$ such that for any $j \in \mathcal{H}$ and $x^* \in \mathcal{X}^*$,*

$$\sum_{i \in \mathcal{S}} f_i\left(\frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} x_j[t]\right) - \sum_{i \in \mathcal{S}} f_i(x^*) \leq \mathcal{O}(1/\sqrt{T}). \quad (8)$$

The proof, omitted here, uses the standard subgradient inequality combined with the Tverberg property of the aggregator: every $y_{ij}[k]$ lies in the convex hull of an honest subset, so the consensus value $v_i[k]$ inherits inputs only from honest objectives.

V. NUMERICAL EXAMPLES

The numerical examples in this section are produced by the open-source Python implementations `ACC22.py` and `ACC23.py` that accompany this exposition. Both provide the worst-case per-recipient “two_faced” Byzantine adversary, brute-force checkers for the respective graph-redundancy conditions, and reproducible CLI demos.

A. Part I: ACC22 – Half-box demo at the origin

We run (2) on $n = 14$ agents with $f = 2$ Byzantine, where each honest agent owns one of four asymmetric half-boxes ($\{x_k \geq 0\}$ or $\{x_k \leq 0\}$ along each axis, three copies per direction) so that $\bigcap \mathcal{X}_i = \{(0,0)\}$ with Hoffman $\mu = 1$. Under the `max_spread` attack (each Byzantine broadcasts the antipode of the honest centroid), $V(0) = 9.55$ decays to $V(80) \approx 1.0 \times 10^{-15}$ – the floating-point floor – with empirical $\rho_{\text{emp}} = 0.638$ at $R^2 = 1.000$. All twelve honest agents converge exactly to the origin. The same setup under `two_faced` (per-recipient) Byzantines yields the identical contraction rate, confirming Theorem 2’s tolerance of the per-recipient adversary.

B. Part I: ACC22 – Polyhedral Corollary-1 setting

Replacing the Box constraints with the corresponding axis half-spaces $\{x : a_k^\top x \leq 0\}$ (twelve honest agents, $f = 1$), the implementation’s brute-force redundancy checker reports the largest k for which $\mathcal{H}$ is k -redundant; for our setup this equals 5. The necessary $2f = 2$ -redundancy is satisfied. The sufficient condition $k > 5$ (from Theorem 2 for $f = 1$) is not strictly met, so (6) predicts $\rho_{\text{pred}} > 1$ – but the algorithm still converges empirically with $\rho_{\text{emp}} = 0.638$, illustrating the *conservativeness* of Theorem 2’s sufficient bound.

C. Part II: ACC23 – Tverberg aggregator at $d = 2, \beta = 1$

On the complete graph $\mathcal{K}_8$ with quadratic objectives sharing a common center (so the network is trivially k -redundant) and the `max_spread` adversary, the iterates of seven honest agents converge from initial spread ≈ 1.8 to machine-precision agreement: at $T = 200$ the honest disagreement is $\approx 2 \times 10^{-19}$, and the honest mean is within $\approx 8 \times 10^{-10}$ of the optimum x^* . The objective gap $f_{\mathcal{H}}(\bar{x}) - f_{\mathcal{H}}(x^*) \approx 2 \times 10^{-18}$ is at the floating-point floor.

D. Part II: ACC23 – $d = 1$ trimmed-mean reduction

With the trimmed-mean closed form, the same setup converges to within $\approx 3 \times 10^{-15}$ of x^* in 100 iterations.

E. Part II: ACC23 – Per-recipient adversary

Replacing `max_spread` with the per-recipient `two_faced` adversary degrades convergence only mildly: the honest mean is within $\approx 1.4 \times 10^{-9}$ of x^* at $T = 200$.

F. Reproducing these experiments

Part I (ACC22). The default box demo with the broadcast `max_spread` adversary, the per-recipient `two_faced` adversary, and the polyhedral Corollary 1 setting are reproduced by:

```
python ACC22.py
python ACC22.py --attack two_faced
python ACC22.py --demo polyhedral
```

Part II (ACC23). The paper aggregator at $d = 2, \beta = 1$ on $\mathcal{K}_8$, the $d = 1$ trimmed-mean reduction, the CWTM comparison baseline, and the per-recipient adversary are reproduced by:

```
python ACC23.py
python ACC23.py --d 1 --n 6
python ACC23.py --aggregator cwtm
python ACC23.py --attack two_faced
```

VI. CONCLUSION

We have presented an integrated exposition of two related Byzantine-resilient algorithms by Zhu, Lin, Velasquez, and Liu [1], [2], each accompanied by a paper-faithful open-source implementation. Read together, the two papers occupy complementary niches in the resilient-multi-agent landscape: ACC22 addresses an inherently geometric question on a topologically simple network, while ACC23 addresses an inherently optimization question on a topologically rich network. Both rely on *redundancy* – of feasibility sets in ACC22, of objective functions in ACC23 – as the crucial ingredient that allows resilience to adversarial inputs.

Several open directions, also flagged in the source papers, are worth emphasizing. ACC22’s sufficient bound (4) is conservative relative to the necessary $2f$ -redundancy; closing this gap is identified as future work, and is corroborated empirically in our experiments where convergence holds well below the sufficient threshold. ACC23’s algorithm becomes computationally expensive for high d or β due to the combinatorial number of subsets and the difficulty of intersecting high-dimensional convex hulls; dimension-independent filtering and block-wise updates are natural avenues for relaxing this. Finally, asynchronous variants of both algorithms with bounded message delays would broaden their applicability to real-world distributed systems.

A complementary algorithm in the same repository, the resilient *average*-consensus protocol of Yuan and Ishii [3] (treated in a separate exposition by the same author), provides a detection-and-recovery alternative for the related average-consensus problem on directed graphs and is recommended as further reading.

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